36121 ARTIFICIAL INTELLIGENCE PRINCIPLES AND APPLICATIONS

Ask in English. Inspect the SQL.

Text-to-SQL Agent turns a plain-English question into read-only SQL a person can check, on SQLite, DuckDB or PostgreSQL.

```
SELECT highlight FROM this_project;
```

- **1 of 8** steps uses a model; it never sees result rows
- **45 of 45** engine tests passed on 3 read-only engines
- **64.6 %** of Spider answers matched, local 9B model
- **+3.6 pts** from sending fewer tables on large databases

1 THE PROBLEM A fluent answer can still be the wrong answer

Published scores fall on enterprise tasks

Spider 1.0	91.2 %	
BIRD	73.0 %	
Spider 2.0 enterprise	21.3 %	69.9 points below Spider 1.0

Share of tasks solved, o1-preview, Spider 2.0 paper (2024); not this project.

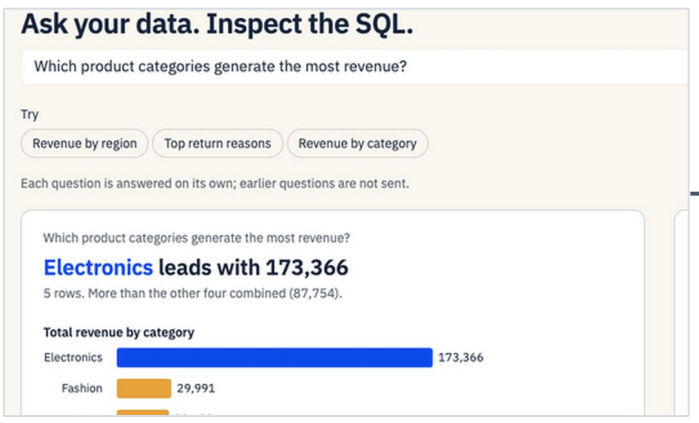
Meaning
"Revenue" is not a column. Gross or net? The schema does not say.

Scale
Real schemas are large: Spider 2.0 databases often exceed 1,000 columns.

Risk
Generated SQL runs on real data, so it must be stopped from changing it.

2 THE DEMO The SQL shows what the answer includes

1 THE APP ANSWERS



2 THE SQL SHOWS HOW

```
SELECT category, SUM(unit_price * quantity
+    * (1 - discount_pct)
  ) AS total_revenue
FROM order_items oi
JOIN products p
  ON oi.product_id = p.product_id
+ JOIN orders o ON oi.order_id = o.order_id
+ WHERE o.order_status = 'Completed'
GROUP BY category
ORDER BY total_revenue DESC;
```

Amber: what completed revenue would add

3 A PERSON DECIDES

Electronics revenue, two definitions

All 220 orders, list price: 173,366

179 completed orders, after discounts: 136,984 (−21 %)

Neither figure is wrong by itself. A person decides which revenue the business means.

3 HOW IT WORKS One step of eight uses a model; the application controls execution

ASK	PREPARE	GENERATE	CHECK AND RUN	ANSWER
1 Ask in plain English	2 Read schema 3 Schema RAG: top 6 tables 4 Build prompt	5 Write SQL or decline; never sees result rows	6 Validate: one read-only query 7 Run with work budget and row cap	8 Show the answer, SQL and why those tables

A query error gets one repair

Database engine by connection string — SQLite, DuckDB, PostgreSQL — Same 15 tests on each

4 SAFETY AND DATA Safety does not rest on the model, and result rows stay out of the prompt

	1 MODEL	2 APPLICATION	3 ENGINE
Write or schema change	Declined 6 of 6	Blocked	Denied
Reading engine internals	Declined 3 of 3	Blocked	Would run
Reading files on disk	—	Blocked	Denied
Runaway or oversized query	—	—	Budget, row cap
Data the schema lacks	Declined 5 of 6	—	—
Valid SQL, wrong meaning	No layer checks this, so the person reads the answer and its SQL		

Enforced and tested · Model behaviour, one run · Not stopped here · — Not this layer's role, or not tested

79 of 608 stored text values reach the prompt

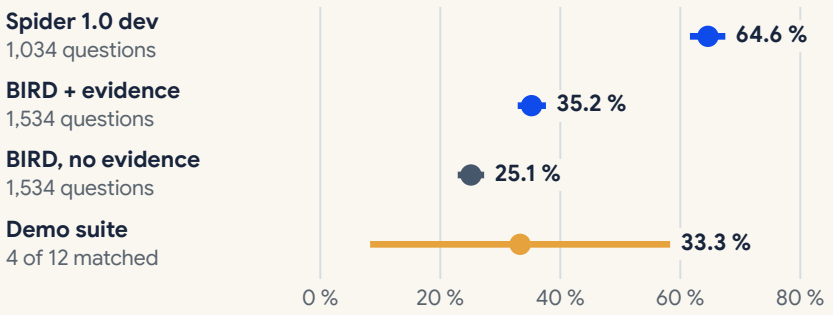
79 in the prompt · 529 not sent

In the prompt	Not in the prompt
Tables, columns, keys	Result rows
79 values as hints	Names, emails
The question	Other values

Hints are real stored values. A local model keeps the prompt on your machine; a hosted one receives it.

5 THE EVIDENCE On large databases, sending fewer tables scored 3.6 points higher

Answers matched, with 95 % intervals



Harder questions score lower: Spider falls from 84.3 % on easy questions to 38.0 % on extra hard ones. Project scorer, not leaderboard scores; local qwen3.5:9b, runs of 10 and 19 October 2026.

Schema sent	Answers matched	Prompt tokens	Time
Every table + hints	25.5 %	8,501	4.0 s
Every table, no hints	26.8 %	6,221	4.2 s
Top 6 + neighbours	29.1 %	2,670	6.0 s

Selection wins until it drops a table

Every table sent, 1,055 questions	32.9 %	26.7 %
A table dropped, 167 questions	5.4 %	18.0 %

+3.6 points [+1.4, +5.8] against every table + hints on 1,222 questions, from the 65- and 34-table databases. Blue is top 6; amber is every table + hints.

6 WHAT WE LEARNED Better retrieval is not better answers; four tests come next

Each fix helps where it aims, and the rest pay

Correct answers vs selected tables; the later scorer left 128 misses

	Lost on 1,094	Gained on 128	
Dense embeddings	−16	+18	+2 net
Send every table	−44	+15	−29 net
+ table summaries	−39	+13	−26 net

- **Business definitions** — Does a few definitions settle "revenue"?
- **Targeted table recovery** — Keep the gains without losing other answers
- **New large schemas** — Does top 6 hold beyond these 4 databases?
- **Hosted model** — Same suites: accuracy, latency, empty SQL

Code github.com/tuannm3812/text-to-sql-agent
Live demo aipa-text-to-sql-agent.streamlit.app

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